

Inflation is Conflict

A Review of Lorenzoni and Werning (NBER 2023)

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Introduction & Motivation

- Traditional models of inflation often rely on money supply (“too much money chasing too few goods”) or economic activity and output gaps.
- While these theories contain elements of truth, they may fall short of isolating the most general mechanism.
- The authors propose an alternative perspective: the most proximate and general cause of inflation is **conflict**.
- Conflict is defined as a disagreement on relative prices between economic agents.
- The paper explores this through two lenses: a stylized model without money, and a flexible macroeconomic framework.

The Mechanism: Conflict + Nominal Rigidities

- How does a disagreement over relative prices become *sustained* inflation? The key friction is **nominal rigidities**.
- If prices were perfectly flexible, conflicting aspirations might just cause an instantaneous leap to a new price level.
- Instead, agents can only occasionally change the subset of prices under their control (staggered pricing).
- Whenever they do, they adjust prices to influence relative prices in their desired direction.
- **The Result:** This staggered adjustment turns a stalemate over relative prices into a continuous, finite rate of generalized inflation.

Part 1: A Stylized Model of Inflation

- To isolate conflict, the authors purposefully stay away from standard macroeconomic models.
- The model features no money, no credit, no interest rates, and no production or employment fluctuations.
- Trade takes place entirely through barter.
- Inflation arises purely from the agents' desire to exercise market power.
- Because output is fixed and money is absent, inflation here cannot be explained by monetary policy or departures from a natural rate of output.

Model Setup: Endowments and Preferences

- The economy consists of two agents (A and B) and two perishable goods (A and B).
- Each agent receives a unit endowment of their respective good each period.
- Preferences within a period are symmetric and given by the utility function $u(c, c')$.
- Here, c denotes consumption of the agent's own good, and c' denotes consumption of the other agent's good.

Model Setup: Staggered Pricing

- Prices are nominal and serve as terms of trade, but there is no physical currency or storage.
- Prices are set in a staggered fashion and remain unchanged for two periods.
- Agent A resets their price in even periods (P_t^*); Agent B resets in odd periods.
- Trade quantities are determined by a take-it-or-leave-it offer made by the “buyer” (the agent who did not reset their price that period).

Solving the Model: The Agent's Problem

- The buyer solves for optimal demand, taking the relative price p as given:

$$V(p) = \max_{c, c'} u(c, c') \quad \text{s.t.} \quad c + pc' = 1$$

- The seller chooses the relative price p^* to maximize their value:

$$p^* \equiv \arg \max_p v(p)$$

$$v(p) \equiv u(1 - D(p), pD(p))$$

- The first-order condition dictates that the seller sets a markup over their marginal rate of substitution:

$$p^* = \frac{\epsilon(p^*)}{\epsilon(p^*) - 1} \cdot \frac{u_c(c, c_-)}{u_{c_-}(c, c_-)}$$

Proposition 1: Constant and Positive Inflation

- The rate of inflation is constant and entirely determined by preferences and endowments.

$$\frac{P_{t+1}^*}{P_t^*} = p^*$$

- **Proposition 1:** Inflation is constant and positive, given by:

$$\frac{P_{t+1}^*}{P_t^*} - 1 = p^* - 1 = \frac{\epsilon(p^*)}{\epsilon(p^*) - 1} \cdot \frac{u_c(c, c_-)}{u_{c_-}(c, c_-)} - 1 \in \left(0, \frac{1}{\epsilon(p^*) - 1}\right)$$

- Intuition: The seller has market power and finds it optimal to sell their goods at a relative price greater than 1.
- They alternate attaining these conflicting aspirations, which drives sustained price increases.
- The authors also that this result also holds in extensions (random buyer seller roles, non-symmetric preferences)

Extensions: Expectations and Matching

- The baseline model features permanent matching, rendering the pricing problem static.
- Extending the model to random matching makes price setting dynamic.
- In this extension, sellers optimize taking expected future inflation Π^e into account:

$$p^*(\Pi^e) \equiv \arg \max_p \left\{ v(p) + \beta V \left(\frac{(\Pi^e)^2}{p} \right) \right\}$$

- Expected inflation directly impacts the chosen relative price, showing that conflict interacts with forward-looking expectations.

$$\frac{p_t^*}{p_{t-1}^*} = p^*(\Pi^e)$$

Part 2: A General Macroeconomic Framework

- The second part of the paper steps away from the stylized barter model.
- It introduces a standard and general framework, fitting many models (staggered prices assumption)
- Introduction of a framework for inflation mechanics based on price and wage setting
- Time is continuous, with a continuum of firms and workers who reset prices and wages with Poisson arrival rates λ_p and λ_w .

$$\pi_t = \lambda_p(p_t^* - p_t) \quad \pi_t^w = \lambda_w(w_t^* - w_t)$$

Aspirations and Conflict

- Behavior is driven by “aspirations” for real wages.
- Firms set their target price based on a desired profit margin aspiration f_t :

$$p_t^* = w_t - f_t$$

- Workers set their target wage based on a desired real wage aspiration g_t :

$$w_t^* = p_t + g_t$$

- **Conflict** exists whenever these two aspirations are incompatible ($f_t \neq g_t$).

Wage and Price Inflation Dynamics

- Letting $\omega_t = w_t - p_t$ denote the current real wage, inflation rates are given by:

$$\pi_t = \lambda_p(\omega_t - f_t)$$

$$\pi_t^w = \lambda_w(g_t - \omega_t)$$

- The dynamic evolution of the real wage is simply the difference between wage and price inflation:

$$\dot{\omega}_t = \pi_t^w - \pi_t$$

Proposition 2: Conflict and Adjustment Inflation

- Define

$$\tilde{\omega} \equiv \alpha f_t + (1 - \alpha)g_t \quad \alpha \equiv \frac{\lambda_p}{\lambda_\omega + \lambda_p}$$

- Price and wage inflation can be decomposed into two distinct components:

$$\pi_t = \Pi_t^C - \alpha \Pi_t^A$$

$$\pi_t^w = \Pi_t^C + (1 - \alpha)\Pi_t^A$$

- Conflict Component:** Driven by the disagreement $g_t - f_t$.

$$\Pi_t^C = (\lambda_p + \lambda_w)\alpha(1 - \alpha)(g_t - f_t)$$

- Adjustment Component:** Determines dynamics of the real wage $\tilde{\omega}_t$.

$$\dot{\omega}_t = \pi_t^w - \pi_t^p = \Pi_t^A = (\lambda_p + \lambda_w)(\tilde{\omega}_t - \omega_t)$$

Propositions 3 & 4: The Centrality of Conflict

- **Proposition 3:** At any point in time, simultaneous price and wage inflation requires conflict ($g_t - f_t > 0$).
- The conflict component acts as the α -weighted average of price and wage inflation. Simultaneous inflation requires conflict.

$$\Pi_t^C = \alpha \pi_t^w + (1 - \alpha) \pi_t$$

- **Proposition 4:** The adjustment component cannot sustain inflation over long periods.
- The long-run average of both price and wage inflation is driven entirely by the average conflict inflation:

$$\bar{\pi} = \bar{\pi}^w = \bar{\Pi}^C$$

- Without conflict there can be no generalized or persistent enough inflation

Forward-Looking Aspirations

- With positive inflation, aspirations for relative prices will not be sustained and will deteriorate over time.
- We consider forward looking agents that have some desired targets $\{\phi_t, \gamma_t\}$ for their relative price and are aware of inflation.
- They adjust their immediate aspirations f_t and g_t to hit these targets on average, over the time their wage or price is fixed.
- This motivates the reset conditions:

$$p_t^* = (\rho + \lambda_p) \hat{E}_t^p \int_0^\infty e^{-(\rho + \lambda_p)s} (w_{t+s} - \phi_{t+s}) ds$$

$$w_t^* = (\rho + \lambda_w) \hat{E}_t^w \int_0^\infty e^{-(\rho + \lambda_w)s} (p_{t+s} + \gamma_{t+s}) ds$$

Aspirations Adjusted by Expected Inflation

- The reset conditions can be recast in terms of aspirations, highlighting the role of inflation expectations:

$$f_t = \hat{E}_t^P \int_0^\infty e^{-(\rho + \lambda_p)s} ((\rho + \lambda_p)\phi_{t+s} - \pi_{t+s}^w) ds$$

$$g_t = \hat{E}_t^w \int_0^\infty e^{-(\rho + \lambda_w)s} ((\rho + \lambda_w)\gamma_{t+s} + \pi_{t+s}) ds$$

- In a steady state with constant targets (ϕ, γ) and expectations $\hat{\pi}$ and $\hat{\pi}^w$:

$$f = \phi - \frac{\hat{\pi}^w}{\rho + \lambda_p}, \quad g = \gamma + \frac{\hat{\pi}^P}{\rho + \lambda_w}$$

- Expected inflation contributes towards disagreement since:

$$g - f = \gamma - \phi + \frac{\hat{\pi}^w}{\rho + \lambda_p} + \frac{\hat{\pi}^P}{\rho + \lambda_w}$$

Inflation Under Rational Expectations

- Imposing rational expectations ($\hat{E}_t^p = \hat{E}_t^w = \mathbb{E}_t$) and differentiating gives:

$$\rho\pi_t = \Lambda_p(\omega_t - \phi_t) + \dot{\pi}_t$$

$$\rho\pi_t^w = \Lambda_w(\gamma_t - \omega_t) + \dot{\pi}_t^w$$

- where $\omega_t = w_t - p_t$ with coefficients $\Lambda_p \equiv \lambda_p(\rho + \lambda_p)$ and $\Lambda_w = \lambda_w(\rho + \lambda_w)$.
- Solving the differential equation for price and wage inflation yields:

$$\pi_t = \Lambda_p \int_0^\infty e^{-\rho s} (\omega_{t+s} - \phi_{t+s}) ds$$

$$\pi_t^w = \Lambda_w \int_0^\infty e^{-\rho s} (\gamma_{t+s} - \omega_{t+s}) ds$$

Proposition 5: Conflict and Adjustment Reprise

- Price and wage inflation can be decomposed into two components:

$$\pi_t = \Pi_t^{C,f} - \alpha^f \Pi_t^{A,f}, \quad \pi_t^w = \Pi_t^{C,f} + (1 - \alpha^f) \Pi_t^{A,f}$$

- The forward-looking conflict component is driven by the present value of differences between ϕ_t and γ_t :

$$\Pi_t^{C,f} = (\Lambda_p + \Lambda_w) \alpha^f (1 - \alpha^f) \int_0^\infty e^{-\rho s} (\gamma_{t+s} - \phi_{t+s}) ds$$

- The forward-looking adjustment component is driven by the present value of differences between the real wage ω_t and target $\tilde{\omega}_t^f$:

$$\Pi_t^{A,f} = (\Lambda_p + \Lambda_w) \int_0^\infty e^{-\rho s} (\tilde{\omega}_{t+s}^f - \omega_{t+s}) ds$$